V2 Rail Guard + Prolabs Flex Sheet (0.2mm)

by Aarav Jaisingh, Prolabs Robotics

Build guide for Bangalore FTC teams | goBILDA FTC StarterBot 2026-27 (BIOBUZZ) | 96mm Hogback, 71mm POLLEN

NOT ROBOT TESTED. We have checked this in CAD only. The 0.2mm sheet is our biggest worry (see Known issues). Please test one first and share what you find so we can improve it together.

Download the files

3D files folder (V2 Shooter 3D files):
https://github.com/Aarav-111/FTC-Bangalore-Resources/tree/main/FTC%20precise%20shooting%20mechanism/V2%20Shooter%203D%20files

FTC Shooter V2 Rail Guard.stl
V2 rail guard, print 1
Download

prolabs_plain_sheet_0p2.stl
Prolabs Flex Sheet, print 1
Download

Scan for the 3D files

Scan for all shooter docs

More docs in the FTC precise shooting mechanism folder:

BOM & 3D printing Guide.md
Open

FTC Shooting and Intake Mechanism Theory.pdf
Open

Shooter Assembly Guide for V2.pdf
Open

1. Open the link ▶ 2. Click the file ▶ 3. Click Download raw file (top right) ▶ Saved

To get everything at once, we open the repo page and click **Code > Download ZIP**.

How to print (beginner friendly)

1. **Open a slicer**

   Use any free slicer: **Cura, PrusaSlicer, Bambu Studio or OrcaSlicer**.

2. **Import the STL**

   File > Import, or drag the STL onto the bed.

3. **Check scale is 100%**

   No unit conversion. V2 should read about **129 x 149 x 88mm**. The sheet should read **88 x 104 x 0.2mm**. If it is tiny or huge, the units are wrong.

4. **Set the orientation**

   V2: lay it sideways on its flat side. **Sheet:** flat on the bed.

5. **Apply the settings**

   Use the settings table below for each part.

6. **Slice and preview**

   The sheet must show **exactly 1 layer** with all holes visible. V2 should have no support.

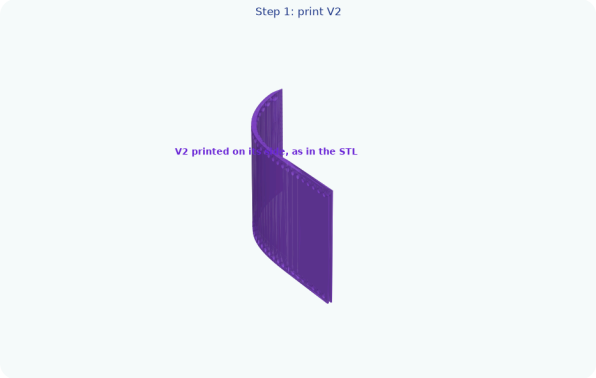
7. **Print and remove**

   Watch the first layer. Peel the sheet off a flexible plate by bending the plate.

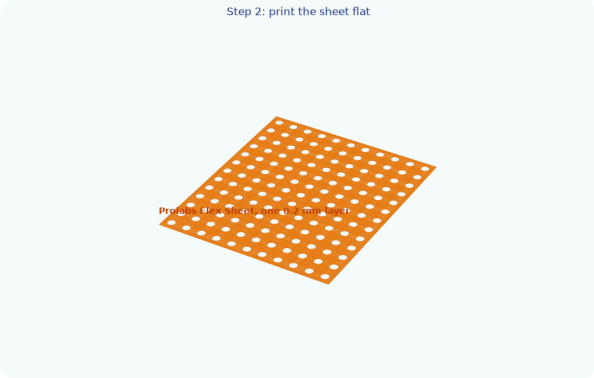
8. **Check dimensions**

   Use calipers. Each size within **+-0.5mm**.

V2 orientation (sideways)



Sheet orientation (flat, one layer)



Setting	V2 rail guard	Prolabs Flex Sheet
Material	PETG or PLA+	PETG
Infill	**100%**	**100%**
Supports	None	None
Layer height	Normal (0.2mm is fine)	**0.2mm, so it is one layer**
Adhesion	5mm brim	No brim; clean, level bed
First layer speed	Normal	**Slow, about 15 to 20 mm/s**
Removal	Normal	Flexible plate, peel gently

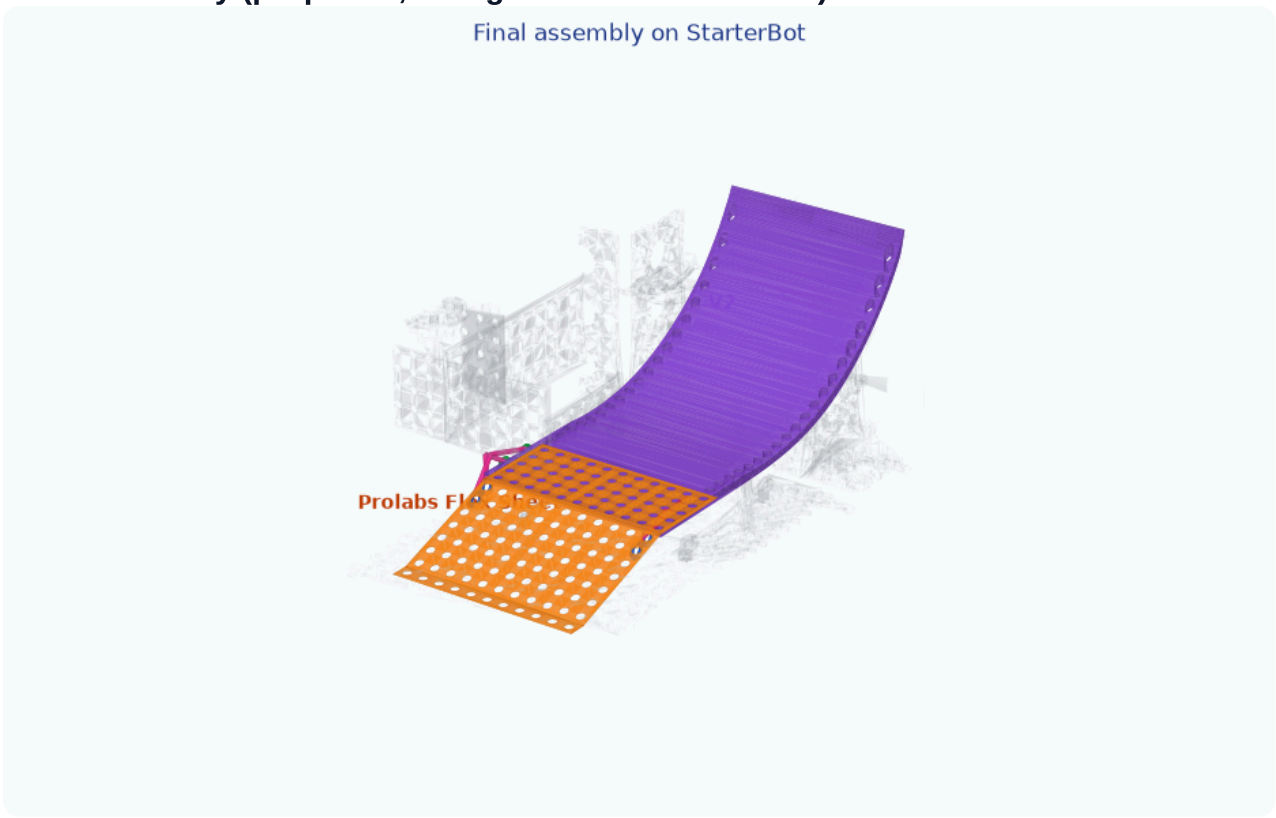
Common problems

Problem	Fix
Sheet will not stick	Clean the bed (soap and water or IPA), re-level, slow the first layer, use a glue stick
Holes closed up	Lower first-layer squish (raise Z offset a little), reduce first-layer flow
Warping (V2)	Keep the brim, raise bed temperature, avoid drafts
Stringing	Dry the PETG, increase retraction, lower nozzle temperature by 5 to 10 C

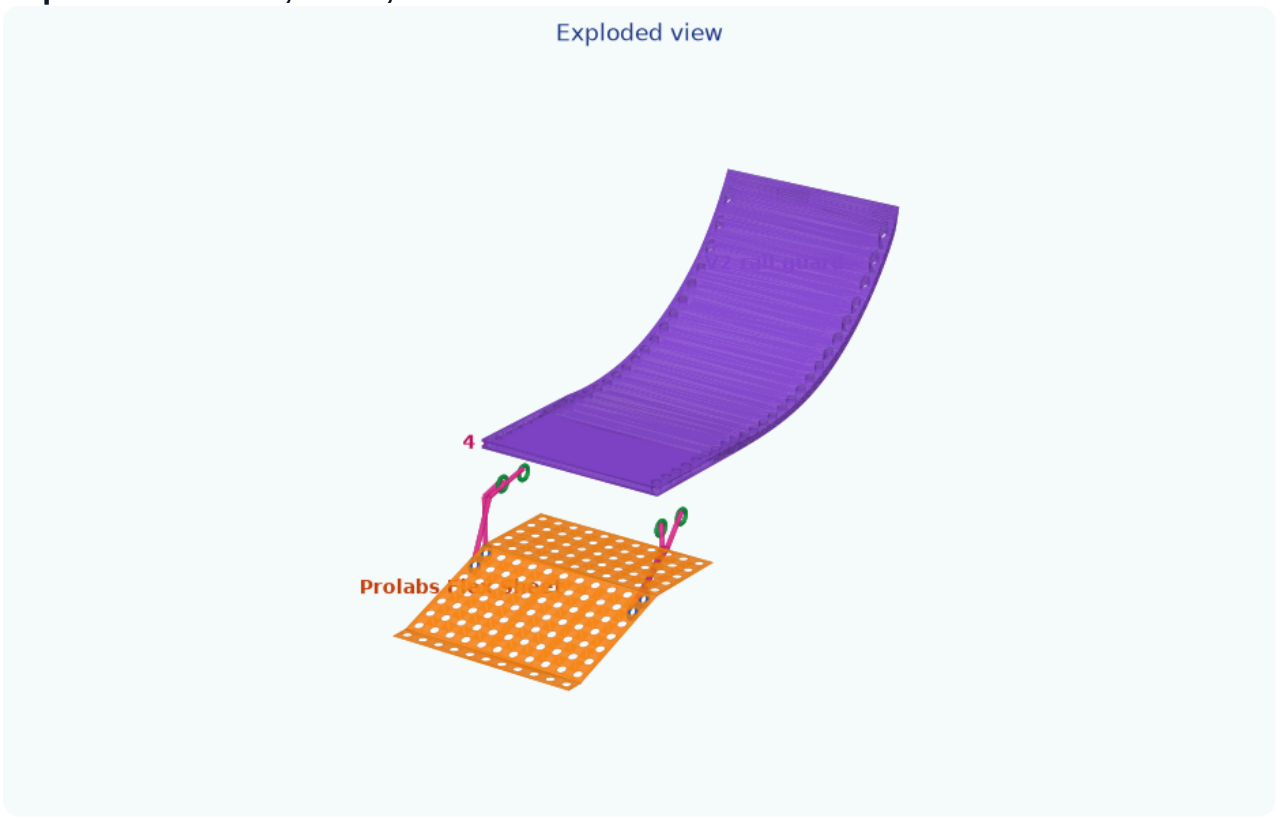
1. How it works

Intake roller ▶ Floor plate ▶ Flex Sheet tail ▶ Flex Sheet slope ▶ Tongue in V2 slot ▶ V2 + Hogback ▶ Shot

Final assembly (purple V2, orange Prolabs Flex Sheet)



Exploded view: V2, sheet, 4 chains



2. Key numbers

Item	Value
V2 size	**129.35 x 148.57 x 88mm**, 55 cm3
Squeeze on 71mm POLLEN	**14.5% peak** (12 to 14.5%), target 10 to 15%
V2 front slot	1.7mm high x 40mm deep
Prolabs Flex Sheet	**88 x 104 x 0.2mm**, 11 x 13 = 143 holes, 4mm on 8mm grid, about 1.9g PETG
Bent shape	40mm tongue, 19 deg slope 56mm, 8mm tail

3. Parts list

Item	Qty	Notes
V2 rail guard (printed)	1	rail_guard_v2.stl
Prolabs Flex Sheet (printed)	1	prolabs_plain_sheet_0p2.stl
Zip ties	**8**	2 per chain x 4 chains
Rubber bands	**4**	all the same size
File or knife	-	Chamfer V2 lip
Calipers, 71mm ball	-	Hand checks

4. Print settings

V2 rail guard

- PETG
- On its side
- No supports
- **100% infill**
- 5mm brim

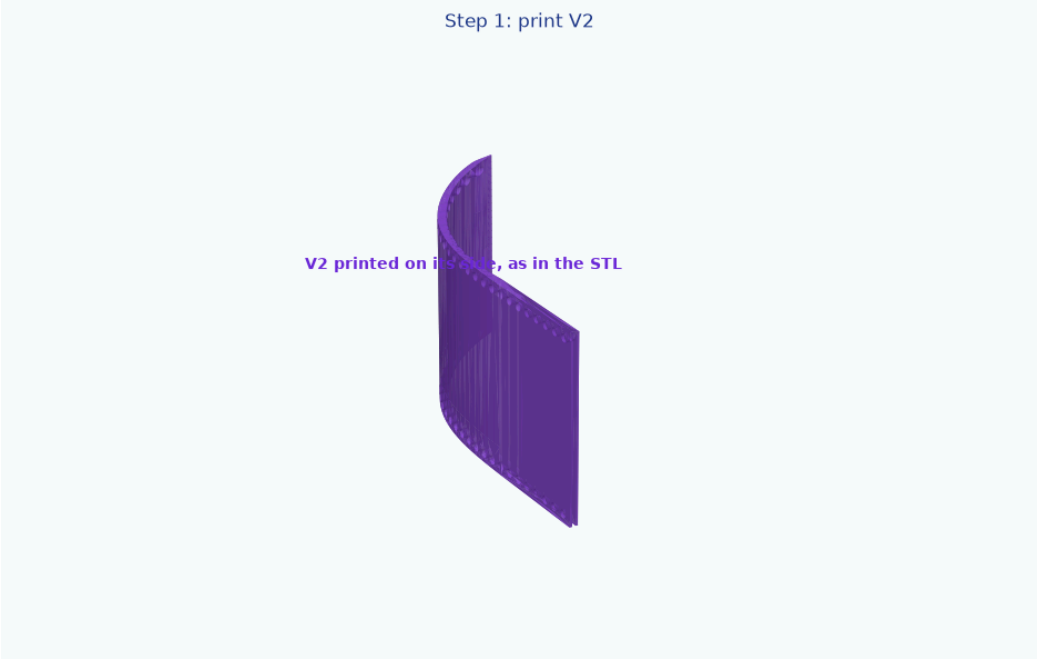
Prolabs Flex Sheet

- PETG, printed flat
- **100% infill**, one 0.2mm layer
- Level the bed carefully
- Slow first layer
- Flexible build plate for removal

5. Step by step

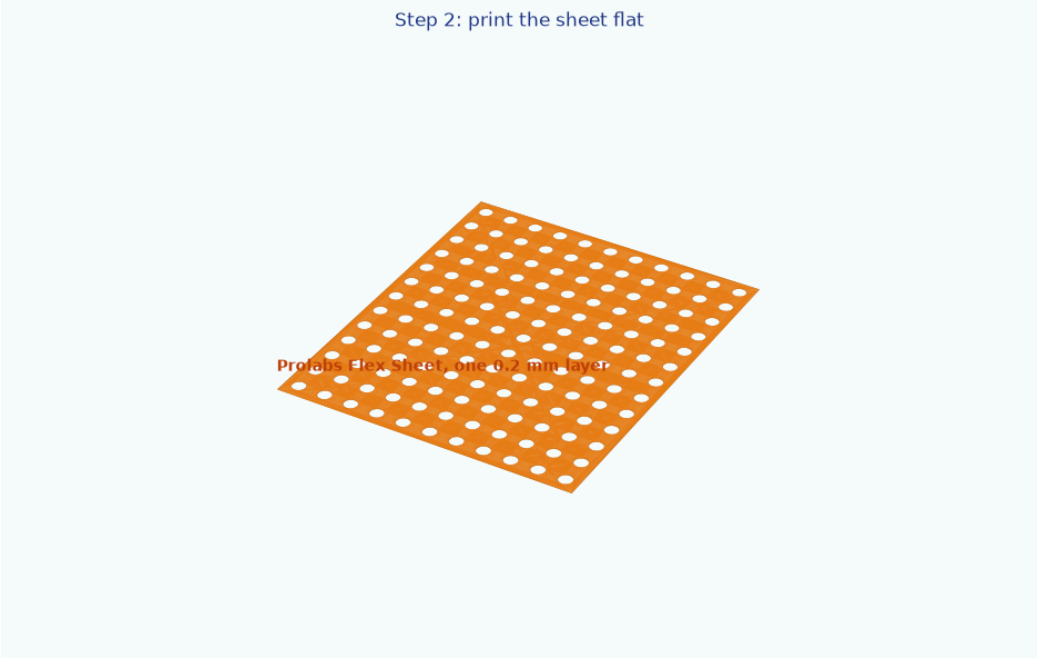
1 Print the V2 rail guard

Print **rail_guard_v2.stl** on its side. **PETG, 100% infill, no supports, 5mm brim.**



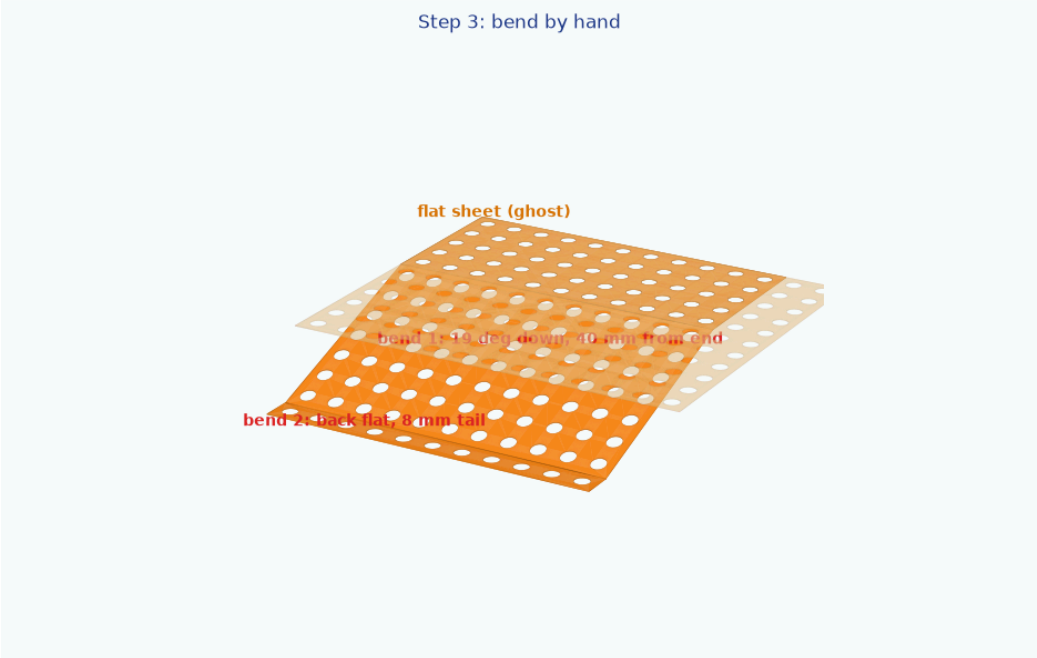
2 Print the Prolabs Flex Sheet

Print **prolabs_plain_sheet_0p2.stl** flat. It is **one 0.2mm layer**, 88 x 104mm, 143 holes of 4mm on an 8mm grid. About 1.9g.



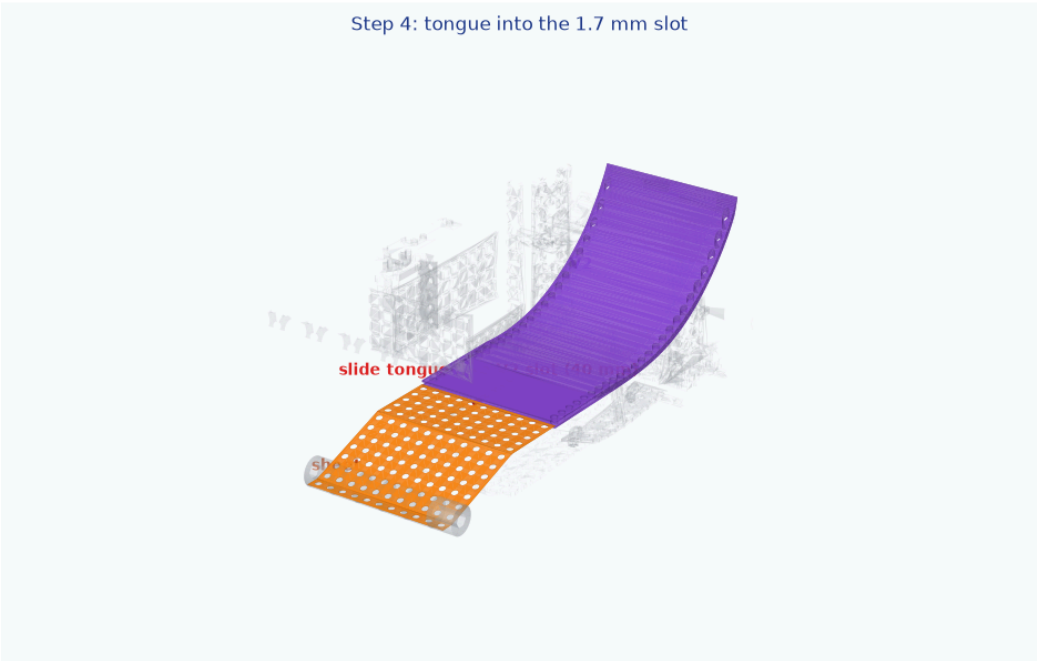
3 Bend it by hand

Bend 1: 19 deg down, 40mm (5 holes) from the tongue end. **Bend 2:** back to flat, 8mm (1 hole) from the tail end. Bend along the webs between hole rows. The sheet is thin, so it bends easily but will not hold the shape on its own.



4 Slide the tongue into the V2 slot

Push the 40mm tongue fully into V2's 1.7mm front slot. The tail runs down to rest on the floor plate.



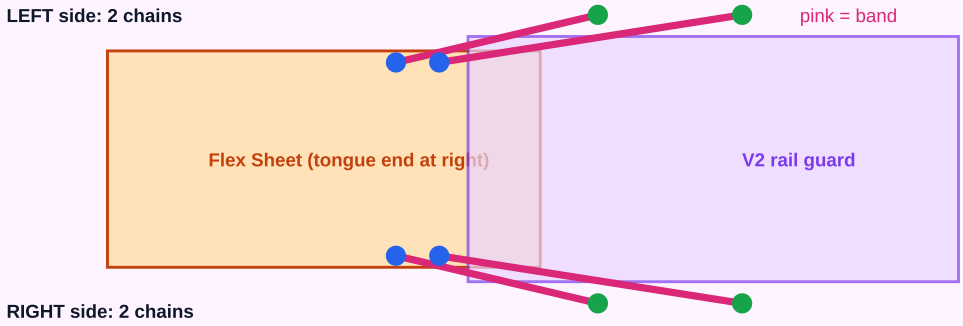
5 Fit the rubber-band chains, both sides

On **each side** make **2 chains**, 4 in total: **sheet hole → zip tie loop → rubber band → zip tie loop → V2 hole**. The bands keep the tongue pulled into the slot.

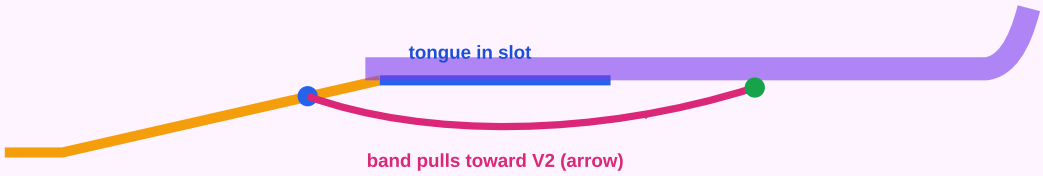
Elastic chain, top view (one chain)



Top view, whole plate: 2 chains per side

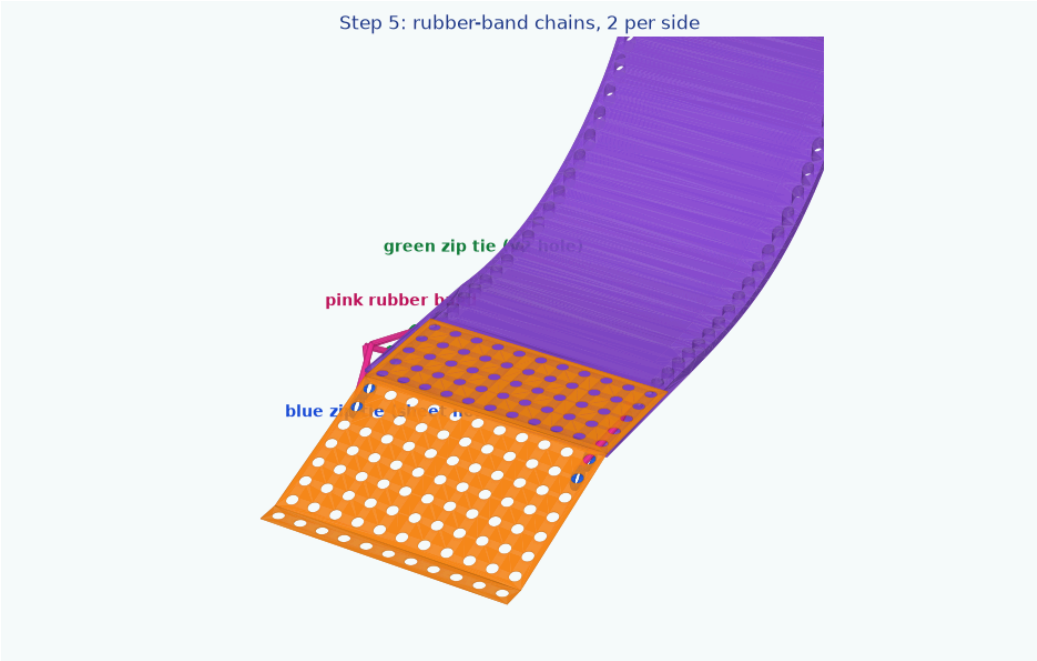


Side view (band pulls tongue into slot)



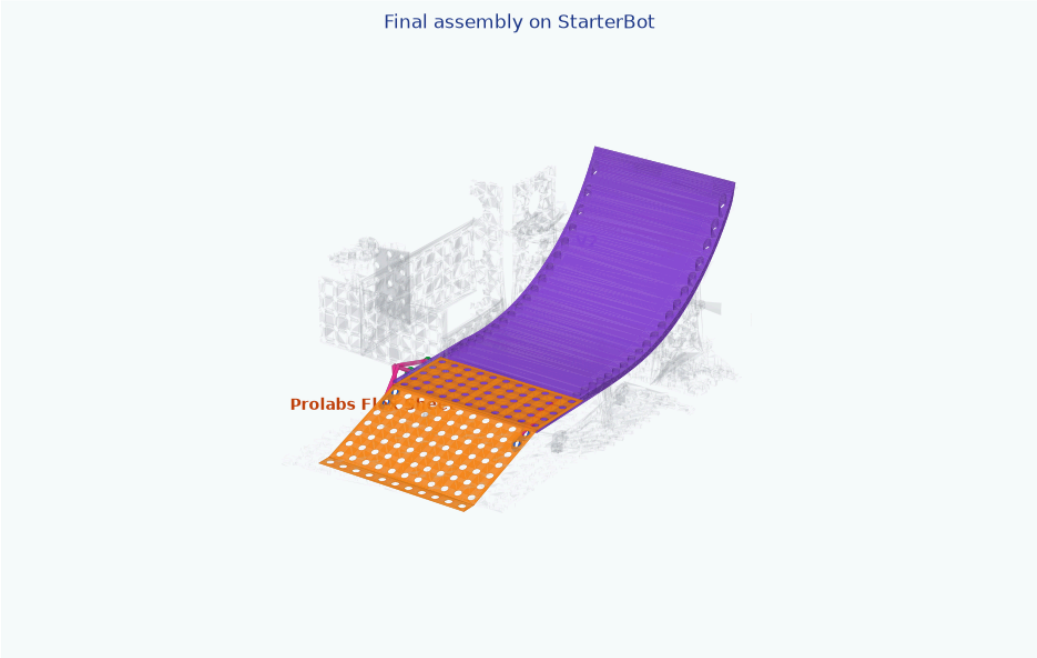
Tips

- **Do not over-tighten** the zip ties. Leave a **loose loop** so the band can stretch.
- Band tension should hold the tongue **fully seated** in the slot.
- Check that **all 4 bands match**. Replace any band that is cracked.
- V2 slot pitch is 7mm and the plate is 8mm, so **pick holes that line up**.
- Trim zip tie tails so they do not touch the ball.



6 Final assembly

Mount V2 on the shooter. Check the tail rests on the floor plate and the sheet runs smoothly from floor to V2.



7 Hand check

Run the checklist below before using it on a robot.

6. Known issues (read before building)

1. **0.2mm sheet sags under the ball**
   A single 0.2mm layer will droop under a 71mm ball and will not hold a bend between tie points. The ball path shape is not controlled. We think a thicker sheet (0.6 to 0.8mm) or ribs would be much safer.

2. **Can tear at the holes**
   If the bands pull hard, the thin sheet can tear between holes. Use light bands and check after every run.

3. **Hole pitch 7mm vs 8mm**
   V2 slot holes are 7.0mm apart (x +-40, y -35.5, -29.1, -21.9, -14.6, -7.6, -0.6). The sheet is on 8mm. Pick holes that line up for the chains.

4. **1.1mm step at V2 lip**
   The sheet sits in the slot under V2's top skin, so the ball meets a 1.1mm step up onto V2. Chamfer the V2 lip edge.

5. **Ball coasts undriven**
   About 131mm with nothing driving it, partly uphill. It needs about 0.7 m/s leaving the intake roller, or a push from the next ball.

6. **Kit part 368 in the path (CAD)**
   An upright plate at y -62 crosses the ball path in CAD. Check on the real robot.

7. **Not tested on a robot**
   Everything here is checked in CAD only.

7. Hand-check checklist

- [ ] V2 size within **+-0.5mm** of 129.35 x 148.57 x 88mm
- [ ] Sheet is one clean layer, no tears at holes
- [ ] Tongue goes fully 40mm into the slot
- [ ] All 4 chains fitted, 2 per side, bands match
- [ ] Pull the sheet gently: it springs back into the slot
- [ ] Roll the 71mm ball by hand: watch how much the sheet sags
- [ ] V2 lip chamfered, no catch
- [ ] No clash with kit part 368 near y -62

8. V2 + Flex Sheet vs V5b

	V2 + Prolabs Flex Sheet	V5b
Parts	2 prints + chains	**1 print**
Material	55 cm3 + 1.9g sheet	227 cm3
Squeeze	14.5%	12.6%
Ramp shape control	Low (0.2mm sags, hand bent)	**Fixed shape**
Team-to-team variation	Higher	**Low**

Honest note: for untested manufacture for many teams, we think V5b is the lower-risk choice. V2 + Flex Sheet is light and cheap to try.

by Aarav Jaisingh, Prolabs Robotics. Feedback from teams is very welcome.